1.8 Counting the Rationals

WORKSHEET

The question

STEP III 2020, Question 8

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Before proving anything, compute the first fifteen or so terms. What do you notice about the ratios u_n / u_{n+1}?

STAGE 1 Odd beats even

Covers: Part (i)

Ask yourself If $(u_k, u_{k+1}) = (a, b)$, what are (u_{2k}, u_{2k+1}) and (u_{2k+1}, u_{2k+2}) in terms of a and b?

Ask yourself A pair (u_n, u_{n+1}) with $n \geq 2$ comes from the pair at position $k = \lfloor n/2 \rfloor$. Treat $n = 2k$ and $n = 2k + 1$ separately.

Pause and try In each case one term of the pair is $u_k + u_{k+1}$. Why is it larger than the other term? Use the fact that every term is positive.

STAGE 2 Consecutive terms are coprime

Covers: Part (ii)

Ask yourself If d divides u_{2k} and u_{2k+1}, what else must it divide? Do the same for u_{2k+1} and u_{2k+2}.

Pause and try So a common factor of a pair is also a common factor of an earlier pair. Take the first pair with a common factor greater than 1, and find a contradiction.

STAGE 3 No pair appears twice

Covers: Part (iii)

Ask yourself Take the earliest repeated pair (a, b). Can it be $(1, 1)$?

Pause and try By part (i), comparing a with b tells you whether the position is odd or even. Use that to step both copies back to an earlier repeated pair.

STAGE 4 Every coprime pair appears

Covers: Part (iv)

Ask yourself Why are $a - b$ and b co-prime? Any common factor of them also divides what?

Pause and try If $a - b$ were followed by b somewhere in the sequence, what pair would appear later on? For the case $a < b$, swap the roles.

STAGE 5 A list of all fractions

Covers: Part (v)

Ask yourself Every positive rational can be written as p/q with p and q co-prime. You need (p, q) to occur as a consecutive pair.

Pause and try Suppose some co-prime pair never occurs, and choose one with the smallest sum. Part (iv) gives a contradiction. Parts (ii) and (iii) then show that f never takes the same value twice.

What's really going on

The Calkin–Wilf tree

Arrange the positive fractions in a tree: put $\frac{1}{1}$ at the top, and give each fraction $\frac{a}{b}$ two children, $\frac{a}{a+b}$ on the left and $\frac{a+b}{b}$ on the right. Reading the tree row by row, left to right, gives exactly the list $f(1), f(2), f(3), \ldots$ from this question (Figure 1.8.1). Stage 3 says every fraction has a unique parent, and Stage 4 says every fraction is in the tree. In 2003 Moshe Newman found that you don't even need the tree: each fraction in the list can be computed from the one before by

$$x_{n+1} = \frac{1}{2\lfloor x_n \rfloor - x_n + 1}.$$

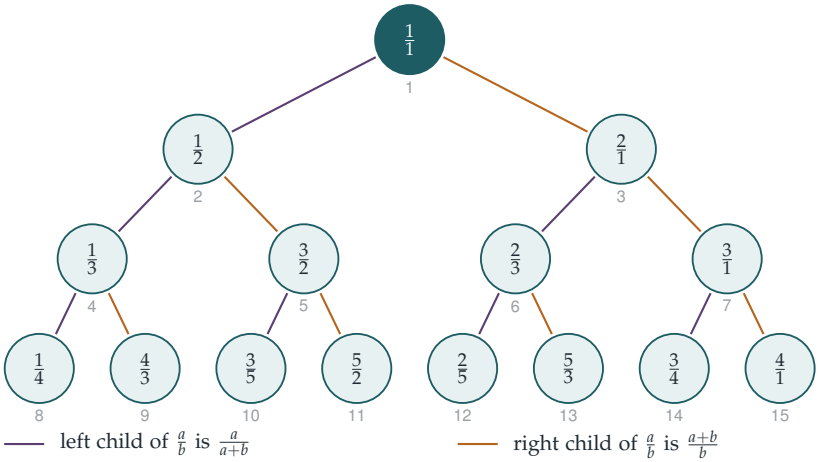


Figure 1.8.1 The first four rows of the Calkin–Wilf tree. Reading row by row, left to right, gives $f(1), f(2), f(3), \ldots$; the small grey numbers are the positions n. Every positive fraction appears exactly once, in lowest terms.

Different sizes of infinity

A set that can be listed like this is called *countable*. The fractions are countable, which is surprising, since they seem so much more numerous than the whole numbers. In your first term at university you will also meet Cantor's *diagonal argument*, which shows that the *real* numbers are not countable: any list of real numbers must miss some. So there really are different sizes of infinity, and the real numbers form a strictly bigger infinity than the rationals. This is one of the most startling results you will meet in a first-year course.

Stern's sequence

The sequence u_n is *Stern's diatomic sequence*, and it has many other lives. For example, u_{n+1} counts the number of ways to write n as a sum of powers of 2 with each power used at most twice. It is closely related to the *Stern–Brocot tree* and to continued fractions, which give the best rational approximations to irrational numbers.

Try this next

1. **Find the fraction.** For which n is $f(n) = \frac{3}{5}$? And $f(n) = \frac{5}{3}$? Use the parent rule to work backwards to $\frac{1}{1}$.

2. **Newman's formula.** Check that Newman's formula produces the first ten fractions of the list, starting from $x_1 = 1$.

3. **Cantor's diagonal argument.** Suppose you had a list of all real numbers between 0 and 1, written as decimals. Construct a decimal that differs from the nth number in the list in its nth digit. Why does this show the list is incomplete? (What care is needed with decimals like $0.0999\ldots = 0.1$?)

4. **Countable unions.** HARDER Show that the set of all pairs (m, n) of positive integers is countable by giving an explicit bijection with the positive integers. (Try $2^{m-1}(2n-1)$.)

Further reading

- **Neil Calkin & Herbert Wilf, "Recounting the rationals"** (*American Mathematical Monthly*, 2000). Four pages, entirely readable.
- **Aigner & Ziegler, *Proofs from THE BOOK***, Chapter 17, "Sets, functions, and the continuum hypothesis", which includes the Calkin–Wilf tree.
- **Ian Stewart & David Tall, *The Foundations of Mathematics*** (Oxford University Press). Countability, Cantor and the real numbers, as in a first-year course.

The full guided solution is in Section 1.8:
https://beyondstep.keithmcnulty.org/ch01/counting-rationals.html